Muon Optimizer: Newton-Schulz Orthogonalization

2 sources

This source consists of a series of handwritten notes concerning a **technical lecture on optimization algorithms**, specifically comparing a recall of the **optimizer recipe** and the **Shampoo algorithm** with the **Muon algorithm**. The notes begin with the general mathematical form of an optimizer, followed by a **detailed breakdown of the Shampoo algorithm** and its underlying matrix operations involving the singular value decomposition (SVD). Crucially, the material introduces Muon, which is described as **Momentum Orthogonalized by Newton Schulz**, as an alternative to the expensive SVD calculation required for semi-orthogonal updates. Muon achieves this by utilizing a **Newton-Schulz iterative process** with specific odd polynomials to approximate the **orthogonalizing step** of replacing singular values with one. The document concludes by summarizing the **Muon update rule** and examining how to choose the polynomial $p(x)$ for the approximation.

The above summar is *just* from uploading the PDF of Lecture #7.

Save to note

I designed this prompt to try to communicate the structure of Cornell notes, but given the capabilities of NotebookLM, I really could have just uploaded an example. This would be an interesting investigation for future use of NotebookLM to generate structured notes.

Today • 8:45 PM

You are an expert in the material covered in the course EECS 182/282A: Deep Learning, in particular, modern optimization algorithms. You are preparing for the final exam. Convert this lecture on MuON (Momentum Orthogonalized) into Cornell Style Notes.
The notes must include:
Left Column: Key Cues/Terms (e.g., 'Newton-Schulz Iteration').
Right Column: Rigorous mathematical definitions and derivations.
Bottom of each Page: A summary to synthesize the main takeaway.

EECS 182/282A: Deep Learning - Lecture 7: MuON (Momentum Orthogonalized)

Key Cues/Terms	Rigorous Mathematical Definitions and Derivations
Lecture Topic	**MuON (M**omentum **O**rthogonalized by **N**ewton Schulz) [1].
Optimization Recipe Goal	The standard optimization goal seeks to minimize the inner product of the gradient and the change in weights (ΔW), constrained by a norm boundary η: $$\text{argmin}_{\|\Delta W\| \leq \eta} \langle \nabla_W \mathcal{L}(W), \Delta W \rangle$$ [2]. The choice of the appropriate norm is crucial [2].
Spectral Norm Choice	If the spectral norm is chosen, and the gradient $G = \nabla_W \mathcal{L}(W)$ is decomposed using SVD, $G = U \Sigma V^T = U_r \Sigma_r V_r^T$, the resulting optimal direction points towards the semi-orthogonal update [2].
Semi-orthogonal Update	The resulting semi-orthogonal update for the weights (key update of Shampoo without accumulation) is: $$W_{t+1} = W_t - \eta \cdot U_r V_r^T$$ [2].
Shampoo (without accumulation)	The update rule for Shampoo (without accumulation) is given mathematically by: $$W_{t+1} = W_t - \eta \cdot (G_t G_t^T)^{-1/4} G_t (G_t^T G_t)^{-1/4}$$ where $G_t = \nabla_W \mathcal{L}(W)$ [3].
SVD Link to Shampoo	If $G_t = U \Sigma V^T$, the complex direction term in the Shampoo update simplifies to the semi-orthogonal matrix, $U_r V_r^T$ [4].
Semi-orthogonal Matrices	A matrix $A \in \mathbb{R}^{m \times n}$ is semi-orthogonal if $AA^T = I_m$ or $A^T A = I_n$ [5].
Benefits of $U_r V_r^T$	The matrix $U_r V_r^T$ has a **condition number of 1**, which ensures that the step taken is uniform in all directions and is **not dominated by the largest singular values** [5].

It feels like the left column is somewhat heavy in text for the purposes of being main ideas, but this may be a limitation of how Gemini can format things.

The lecture notes were a little more explicit about Shampoo vs. Shampoo with accumulation, which NotebookLM seems to have decided not to include in the notes.

The point about *why* we don't want the step to be dominated by the largest singular values is subtle, and the notes do not address this at all.

MuON Key Idea 1: RMS-RMS Norm	MuON uses the RMS-RMS norm constraint, leading to: $$\text{argmin}_{\|\Delta W\|_{\text{RMS}-\text{RMS}} \le \eta} \langle \nabla_W \mathcal{L}(W), \Delta W \rangle$$ 5. This is equivalent to constraining the L2 norm by $\eta \cdot \sqrt{d_{\text{out}}/d_{\text{in}}}$ 5.
MuON Optimal Direction ΔW^+	The resulting optimal update direction is: $$\Delta W^+ = -\eta \cdot \sqrt{\frac{d_{\text{out}}}{d_{\text{in}}}} \cdot U_r V_r^T$$ 5.
Computational Bottleneck	Computation of the direction UV^T is expensive because it requires performing a **Singular Value Decomposition (SVD)** 6.
MuON Key Idea 2: Solution	MuON solves the expense issue by recognizing that (1) approximately correct directions are sufficient, and (2) using **Newton-Schulz Iterations** to find an approximation 6.

Summary Page 1: MuON leverages the optimal update direction $U_r V_r^T$, which provides steps with a condition number of 1, promoting uniform convergence across all directions, an improvement over methods sensitive to extreme singular values 5. This direction is derived by minimizing the gradient projection under constraints like the RMS-RMS norm 5. Because calculating $U_r V_r^T$ requires expensive SVD, MuON substitutes this step with an efficient approximation technique: Newton-Schulz iteration 6.

This summary is OK, but seems to skip over the connection to Shampoo which was emphasized somewhat heavily in the lecture. Overall it is correct, but doesn't seem fully comprehensive due to this ommission.

Key Cues/Terms	Rigorous Mathematical Definitions and Derivations
Newton-Schulz Goal	The objective is to approximate the transformation from the gradient $U\Sigma V^T$ to the desired semi-orthogonal direction UV^T 6. This means replacing all singular values in the diagonal matrix Σ with 1 6.
Odd Polynomials Property	Newton-Schulz utilizes **odd polynomials** $p(X)$ which commute with the SVD 7. An odd polynomial is defined as: $$p(X) = a_0 X + a_1 X X^T X + a_2 (X X^T)^2 X + \dots$$ 7.
SVD Commutation	If X has SVD $X = U\Sigma V^T$, then applying the odd polynomial yields: $$p(X) = U p(\Sigma) V^T$$ 7. This property allows applying the transformation p only to the singular values (Σ) without needing to recompute the singular vectors (U, V) 7.
Example Iteration Polynomial	A typical polynomial used for iterative scaling is: $$p(x) = \frac{3}{2}x - \frac{1}{2}x^3$$ 8. Iteratively applying $p(x)$ to a singular value σ causes $p^n(\sigma) \to 1$ as the number of iterations $n \to \infty$ 8.
Prerequisite for Convergence	For the iterative scaling using $p(x)$ to converge to 1, the singular values must be strictly contained within the range $\sigma \in (0, 1)$ 9.
Normalization Method	To ensure singular values are in the required range, the gradient G must be normalized by the Frobenius norm: $$\frac{G}{\|G\|_F}$$ 9.
MuON Algorithm (Steps)	MuON combines momentum with the Newton-Schulz orthogonalization 1:

The lecture notes used an example to help illustrate why scaling by the Frobenius norm is used here, but NotebookLM can't seamlessly integrate something like that, which is certainly a limitation.

Step 1: Momentum Accumulation	$$B_t = \mu \cdot B_{t-1} + \nabla_W \mathcal{L}(W)$$ 1.
Step 2: Orthogonalization	$$O_t = \text{NewtonSchulz}(B_t)$$ 1.
Step 3: Weight Update	$$W_t = W_{t-1} - \eta \cdot O_t$$ 1.
Polynomial Order Choice	The general form of the required odd polynomial is $p(x) = ax + bx^3 + cx^5 + \dots$ 1. While **higher-order polynomials may converge faster**, each step of the optimization becomes **more expensive** 1.

This summary seems better than the previous one to me in that it captures the main ideas in this section. I'm not sure how it decided how to break up the notes into "pages", which is usually arbitrary when taking Cornell notes by hand on paper.

Summary Page 2: MuON utilizes the Newton-Schulz method to efficiently approximate the semi-orthogonal matrix UV^T 6. This approximation relies on the property that odd polynomials commute with the SVD, allowing singular values to be iteratively pushed toward 1 (e.g., using $p(x) = \frac{3}{2}x - \frac{1}{2}x^3$) without explicitly calculating the singular vectors 7 8. Before applying the iteration, the gradient must be normalized by the Frobenius norm to ensure singular values fall within the necessary convergence range of $(0, 1)$ 9. The final MuON algorithm integrates momentum accumulation (B_t) with this orthogonalization step (O_t) 1.

On page 1 of your Cornell notes, you went over how Muon uses some ideas from Shampoo. Can you clarify how Muon and Shampoo algorithms differ in their update rules for optimization? Please refer to formulas from the notes or from alternative sources.

The MuON and Shampoo algorithms share the foundational concept of attempting to orthogonalize the gradient direction, primarily differing in how they compute this direction and whether they incorporate momentum or accumulated statistics.

As discussed, the notes are less extensive than the provided lecture handwritten slides in explaining the connectoin between Shampoo and Muon.

Shared Foundation: The Semi-Orthogonal Update

Both MuON and Shampoo aim for an update direction based on the semi-orthogonal matrix $U_r V_r^T$, derived from the Singular Value Decomposition (SVD) of the gradient $G = \nabla_W \mathcal{L}(W)$ 1.

The optimal direction when minimizing the gradient-step inner product constrained by the spectral norm ($\|\Delta W\|_\Sigma \leq \eta$) results in the semi-orthogonal update 1:

$$W_{t+1} = W_t - \eta \cdot U_r V_r^T$$

This specific update is described as the **"Key update of Shampoo (without accumulation)"** in the sources 1. This direction is desirable because $U_r V_r^T$ has a condition number of 1, which ensures a uniform step in all directions and prevents the update from being dominated by the largest singular values 2.

Shampoo Update Rule

Shampoo (without accumulation) explicitly uses matrix roots based on the gradient G_t 3.

The update rule for **Shampoo without accumulation** is:

$$W_{t+1} = W_t - \eta \cdot (G_t G_t^T)^{-1/4} G_t (G_t^T G_t)^{-1/4}$$

3

This mathematically complex form simplifies to the desired semi-orthogonal direction $U_r V_r^T$ 4. Specifically, if $G_t = U \Sigma V^T$, the scaling term simplifies as follows:

$$(G_t G_t^T)^{-1/4} G_t (G_t^T G_t)^{-1/4} = U_r V_r^T$$

4

The standard (or "full") **Shampoo update** introduces accumulated statistics (L_t and R_t, which are approximations of the second-moment information $G_t G_t^T$ and $G_t^T G_t$) 3:

$$L_t = L_{t-1} + G_t G_t^T$$

$$R_t = R_{t-1} + G_t^T G_t$$

$$W_{t+1} = W_t - \eta \cdot L_t^{-1/4} G_t R_t^{-1/4}$$

[3]

MuON Update Rule

MuON (Momentum Orthogonalized by Newton Schulz) differs from Shampoo in three key ways: its normalization source (RMS-RMS norm), its use of momentum, and its efficient approximation technique (Newton-Schulz).

1. **Optimal Direction (RMS-RMS Norm):** MuON is explicitly derived by constraining the RMS-RMS norm, which results in a similar semi-orthogonal direction, but scaled by a factor related to the dimensions of the weight matrix W [2]:

$$\Delta W^+ = -\eta \cdot \sqrt{\frac{d_{\text{out}}}{d_{\text{in}}}} \cdot U_r V_r^T$$

2. **Algorithm Structure and Approximation:** MuON avoids the direct calculation of matrix roots (as seen in Shampoo) or SVD (which is necessary to calculate $U_r V_r^T$) [5]. Instead, it integrates momentum (B_t) and uses **Newton-Schulz iteration** (NewtonSchulz) to approximate the orthogonal direction UV^T from the momentum [6].

The MuON algorithm follows these steps [6]:

- **Step 1: Momentum Accumulation:**

$$B_t = \mu \cdot B_{t-1} + \nabla_W \mathcal{L}(W)$$

- **Step 2: Orthogonalization:**

$$O_t = \text{NewtonSchulz}(B_t)$$

(This step approximates the scaling/orthogonalization $B_t \to UV^T$) [5].

- **Step 3: Weight Update:**

$$W_t = W_{t-1} - \eta \cdot O_t$$

It feels like this is not really connecting the Shampoo algorithm to the Muon algorithm, but rather regurgitating equations from both separate parts of the paper. This may be due to the system being designed to prioritize citations as it was designed as a RAG system.

Summary of Differences

Feature	Shampoo (Without Accumulation)	MuON
Optimization Goal	Based on spectral norm constraint [1], requiring computation of matrix root of gradient outer products [3].	Based on RMS-RMS norm constraint [2].
Input for Direction	The current gradient, G_t [3].	The accumulated momentum, B_t [6].
Direction Calculation	Directly calculates $\left(G_t G_t^T\right)^{-1/4} G_t \left(G_t^T G_t\right)^{-1/4}$ (equivalent to $U_r V_r^T$) [3] [4].	Approximates the semi-orthogonal matrix using iterative application of odd polynomials (Newton-Schulz iteration) [5] [6].
Update Form	$W_{t+1} = W_t - \eta \cdot [\text{Direction based on } G_t]$ [3].	$W_t = W_{t-1} - \eta \cdot [\text{Orthogonalized Momentum } O_t]$ [6].

This is a helpful table, but not in the Cornell notes style, so if I were copying the notes I'd have to reformat these.

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On page 2, can you explain why the Normalization Method uses the Frobenius norm? This would be helpful to include in the notes column.

The Normalization Method in MuON utilizes the Frobenius norm because the **Newton-Schulz iteration requires its input singular values to be strictly within the range** $(0, 1)$ for the polynomial approximation to converge to 1 [1].

When approximating the desired semi-orthogonal matrix UV^T from the gradient G (or momentum B_t), the goal is to transform the singular values (Σ) within $U\Sigma V^T$ so that they all become 1 [2].

The convergence of the iterative process, such as applying $p(x) = \frac{3}{2}x - \frac{1}{2}x^3$ repeatedly, is guaranteed only if the initial singular values σ are bounded between 0 and 1 ($\sigma \in (0, 1)$) [1].

Therefore, to meet this strict convergence prerequisite, the gradient G (or accumulated momentum B_t) must be normalized by its **Frobenius norm** ($\|G\|_F$), resulting in the normalized matrix $\frac{G}{\|G\|_F}$ [1]. This scaling ensures that the magnitude of the largest singular value, and consequently all singular values, is less than or equal to 1, fitting the necessary convergence criterion.

In the actual lecture, we had a plot that we used to visualize an example. Since we are lacking that here, I wanted to probe to see if it could explain or if it would try to use some sort of example.

Here is how this information would be detailed in your Cornell Notes (Page 2, Right Column):

Key Cues/Terms	Rigorous Mathematical Definitions and Derivations
Prerequisite for Convergence	The iterative scaling process using the odd polynomial $p(x)$ (e.g., $p(x) = \frac{3}{2}x - \frac{1}{2}x^3$) requires that all singular values σ of the input matrix fall strictly within the range $\sigma \in (0, 1)$ for $p^n(\sigma) \to 1$ as $n \to \infty$ 1 .
Normalization Method (Frobenius Norm Rationale)	To guarantee that singular values are contained within the required convergence range $(0, 1)$, the gradient G (or momentum B_t) must be scaled by the Frobenius norm: $$\frac{G}{\|G\|_F}$$. Normalizing by the Frobenius norm ensures that the magnitude of the matrix is bounded, satisfying the prerequisite for the Newton-Schulz iteration 1 .

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So based on the algorithm you wrote out on Page 2, do you calculate B_t, O_t, and W_t for each step? Can you explain the purpose of each of these terms?

Yes, based on the MuON algorithm steps detailed, $\mathbf{B_t}$, $\mathbf{O_t}$, and $\mathbf{W_t}$ are calculated sequentially at each optimization step t.

I wanted to get more insight on the Muon algorithm itself since the presentation was somewhat disjoint.

The MuON algorithm combines momentum accumulation with orthogonalization using Newton-Schulz iteration to determine the weight update 1 .

Here is an explanation of the purpose of each term in the MuON update sequence:

Term	Calculation Step	Purpose/Explanation
B_t	**Step 1: Momentum Accumulation** $B_t = \mu \cdot B_{t-1} + \nabla_W \mathcal{L}(W)$ 1	**Accumulated Momentum:** B_t is the accumulated momentum term, which is a running average of the gradient of the loss ($\nabla_W \mathcal{L}(W)$) at the current step, mixed with the prior momentum B_{t-1} using the decay factor μ 1 . This term captures historical gradient information, providing stability and acceleration to the optimization process, similar to standard momentum techniques.
O_t	**Step 2: Orthogonalization** $O_t = \mathrm{NewtonSchulz}(B_t)$ 1	**Orthogonalized Direction:** O_t is the resulting direction matrix after applying the **Newton-Schulz iteration** to the accumulated momentum B_t 1 . The goal of this process is to transform the momentum $B_t = U\Sigma V^T$ into the semi-orthogonal direction UV^T by approximating the replacement of all singular values in Σ with 1 1 2 . This provides a desirable step direction with a condition number of 1, ensuring a uniform step across all dimensions 3 .
W_t	**Step 3: Weight Update** $W_t = W_{t-1} - \eta \cdot O_t$ 1	**Updated Weights:** W_t represents the model weights at the current time step 1 . The weights are updated by subtracting the learning rate (η) multiplied by the calculated orthogonalized direction (O_t). This step is the core mechanism for minimizing the loss function based on the specialized, orthogonalized step direction 1 .

In essence, MuON takes the momentum (B_t) and uses the Newton-Schulz method to **orthogonalize** it (O_t), ensuring the optimization step taken to update the weights (W_t) is highly effective by being uniform in all directions 1 3 . This avoids the computational expense of full SVD required to find the true UV^T direction 2 .

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You stated that singular values must be strictly in $(0,1)$. What happens if the gradient matrix is rank deficient (i.e., has a singular value $\sigma = 0$)?

The Newton-Schulz iteration used in MuON requires that all singular values of the input matrix (the normalized momentum $\frac{B_t}{\|B_t\|_F}$) be **strictly contained within the range** $(0, 1)$ for the process to converge to 1 1 .

I wanted to test an edge case to see if NotebookLM would go beyond the explicit equations in the uploaded PDF, which it correctly responds to.

If the gradient matrix (and thus the accumulated momentum B_t) is rank deficient, it means it contains at least one singular value $\sigma = 0$.

Here is what happens in the MuON algorithm when $\sigma = 0$:

1. **Violation of Strict Constraint:** The presence of $\sigma = 0$ **violates the strict prerequisite** ($\sigma \in (0, 1)$) needed for the Newton-Schulz iteration to converge toward the value of 1 1 .

2. **Behavior of the Odd Polynomial:** When applying the odd polynomial $p(x)$ used in the iteration, such as $p(x) = \frac{3}{2}x - \frac{1}{2}x^3$ 2 , to a singular value of zero:

$$p(0) = \frac{3}{2}(0) - \frac{1}{2}(0)^3 = 0$$

The iteration leaves the zero singular value unchanged. If $p^n(\sigma)$ starts at 0, it will remain 0.

3. **Impact on Orthogonalization:** The MuON algorithm aims to transform $U\Sigma V^T$ into the semi-orthogonal direction UV^T by replacing all singular values with 1 3 . For a rank-deficient direction, the Newton-Schulz approximation will successfully push the non-zero singular values toward 1, but it will leave the zero singular value as 0.

In practical terms, this result is acceptable because the zero singular value corresponds to the null space of the gradient (a direction where the gradient has no component). The optimal semi-orthogonal direction derived from the optimization recipe, $U_r V_r^T$, naturally involves only the components corresponding to the non-zero rank r of the matrix 4 5 . By keeping the output corresponding to $\sigma = 0$ at zero, the MuON update direction O_t does not introduce movement in directions where the momentum lacks information.

Key Cues/Terms	Rigorous Mathematical Definitions and Derivations
Rank Deficient Matrix ($\sigma = 0$)	If a singular value $\sigma = 0$, this violates the **strict requirement** that singular values must be $\sigma \in (0, 1)$ for the Newton-Schulz process to converge to 1 1 .
Effect on Iteration	Applying the odd polynomial $p(x)$, such as $p(x) = \frac{3}{2}x - \frac{1}{2}x^3$, to $\sigma = 0$ results in $p(0) = 0$ 2 . The iteration **does not converge to 1** in this direction; it remains 0.
Optimization Implication	This outcome is functionally appropriate, as the zero singular value corresponds to the null space of the gradient/momentum. The optimal semi-orthogonal direction $U_r V_r^T$ only uses the rank r components, naturally excluding directions where the gradient is zero 4 .